Project Title: Integrated Student Enrollment & Course Management System (ISE-CMS)

Institution: Uganda Martyrs University – Faculty of Science

Course Unit: Web Based Programming

Submission Date:8 May 2026

Development Group Members:

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System Overview

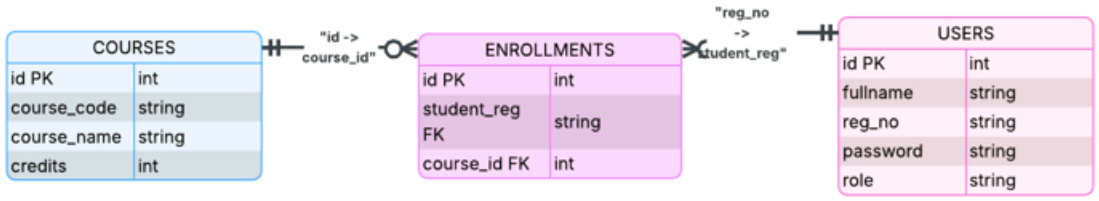
This is a dynamic web application designed to solve the challenges of manual academic course registration. It provides a centralized digital environment where students can securely manage their academic progress.

Key Functional Modules:

- **Authentication Gateway:** A secure login system that differentiates between **Student** and **Admin** roles using session-based validation.
- **Dynamic Course Catalog:** A searchable database of available course units, allowing students to filter options by Course Name or Code.
- **Enrollment Engine:** A logical layer that processes registration requests and prevents duplicate entries for the same course.
- **Administrative Control Panel:** A protected interface for staff to perform **CRUD operations** (Create, Read, Update, Delete) on the course catalog to keep the curriculum up to date.

How the system works

- **User Registration:** * The user enters their Full Name, Reg No, and Password.
    - **Logic:** The system checks if the Reg No already exists. If not, it uses saves the record to the `users` table.
- **Authentication (Login):** * The user logs in with their credentials.
    - **Logic:** The system starts a `$_SESSION`, fetches the user's **Role**, and redirects them to the correct dashboard (Student or Admin).
- **Course Enrollment (Student):** * The student searches for a course using the search bar (**SQL LIKE query**).
    - They click "Enroll." The system runs a **Pre-check** to ensure they aren't already registered, then updates the `enrollments` table.
- **Course Management (Admin):** * The Admin views the master list.
    - They can click "Delete" on any record. The system sends the ID to a deletion script that removes the row from the `courses` table.



CHALLENGES

Fear of Breaking the System: There is a genuine fear that if you change one line of code in the database connection, you might delete everything so it kind of made things hard because of pressure.

Establishing a connection to the database was hard. Root user was locked out and couldn't gain admin privileges

Even with help of a tool some code is just too complicated to comprehend

LESSONS

we learned that the most important part of coding happens before you even touch the keyboard. In the beginning, we tried to write PHP code immediately and got lost. we learned that sketching the **Logic Flow** on paper (Pseudo-code) saves hours of frustration.

We learned that the database is the core of a given system

PORTAL LOGIN

USERNAME / REG NO

e.g. 2024-B071-22936

PASSWORD

••••••••

LOG IN

Don't have an account? Register here

Create Account

Student Registration Portal

FULL NAME

Enter Full Name

REGISTRATION NUMBER

23344455

PASSWORD

•••••

Welcome, lawrence tusimme

Reg No: 23344455

Logout

Search (e.g. SQL, Java, PHP)...

Filter Courses

BIT1101 — 4 CU

Introduction to ICT

Dr. Okello

Already Enrolled

BIT1202 — 3 CU

Computer Networks

Eng. Musoke

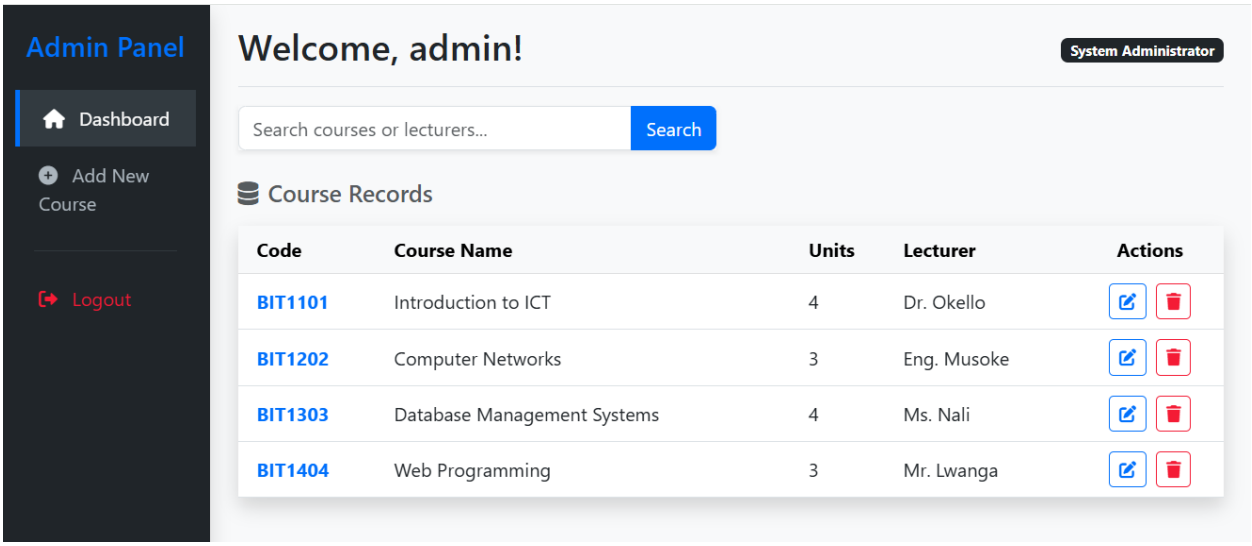
Enrol Me

BIT1303 — 4 CU

Database Management Systems

Ms. Nali

Enrol Me


Admin Panel
Dashboard
Add New Course
Logout
Welcome, admin!
System Administrator
Search courses or lecturers...
Search
Course Records
Code
Course Name
Units
Lecturer
Actions
BIT1101
Introduction to ICT
4
Dr. Okello
BIT1202
Computer Networks
3
Eng. Musoke
BIT1303
Database Management Systems
4
Ms. Nali
BIT1404
Web Programming
3
Mr. Lwanga